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Abstract: Prostate cancer is a highly incident malignant cancer among men. Early detection of prostate cancer is necessary for deciding whether a patient should receive costly and invasive biopsy with possible serious complications. However, existing cancer diagnosis methods based on data mining only focus on diagnostic accuracy without paying attention to the interpretability of the diagnosis model that is necessary for helping doctors make clinical decisions. To take both accuracy and interpretability into consideration, we propose a stacking-based ensemble learning method for simultaneously constructing diagnostic model and extracting interpretable diagnostic rules. For this purpose, a multi-objective optimization algorithm is adopted to maximize the classification accuracy and minimize the ensemble complexity for model selection. The base learners, which are decision trees, are combined using the random forest classifier-based stacking model. Empirical results on real-world data from the General Hospital of PLA have shown that the classification performance of the proposed method outperforms several state-of-the-art methods in terms of classification accuracy, sensitivity and specificity. Moreover, we demonstrate that several diagnostic rules extracted from the constructed ensemble learning model are accurate and interpretable.

Dear Editor,

We submit our paper titled "Stacking-based ensemble learning of decision trees for interpretable prostate cancer detection" to your journal "Applied Soft Computing".

This manuscript is our original work and has not been published or has it been submitted simultaneously elsewhere. All authors have checked the manuscript and have agreed to the submission.

If you have any question about our work, please let us know as soon as possible.

Best regards,

Dujuan Wang
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Highlights

- We propose a stacking-based interpretable selective ensemble learning method.
- We select ensemble models with accuracy and complexity under consideration.
- We combine selected effective models by random forest-based stacking.
- The proposed method is more accurate and interpretable in prostate cancer detection.
- We extract a few of effective diagnostic rules for clinical decision support.

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Stacking-based ensemble learning of decision trees for interpretable prostate cancer detection

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Abstract

Prostate cancer is a highly incident malignant cancer among men. Early detection of prostate cancer is necessary for deciding whether a patient should receive costly and invasive biopsy with possible serious complications. However, existing cancer diagnosis methods based on data mining only focus on diagnostic accuracy without paying attention to the interpretability of the diagnosis model that is necessary for helping doctors make clinical decisions. To take both accuracy and interpretability into consideration, we propose a stacking-based ensemble learning method for simultaneously constructing diagnostic model and extracting interpretable diagnostic rules. For this purpose, a multi-objective optimization algorithm is adopted to maximize the classification accuracy and minimize the ensemble complexity for model selection. The base learners, which are decision trees, are combined using the random forest classifier-based stacking model. Empirical results on real-world data from the General Hospital of PLA have shown that the classification performance of the proposed method outperforms several state-of-the-art methods in terms of classification accuracy, sensitivity and specificity. Moreover, we demonstrate that several diagnostic rules extracted from the constructed ensemble learning model are accurate and interpretable.

Key words: Prostate cancer detection; Ensemble learning; Stacking; Rule extraction; Multi-objective optimization

1 Introduction

Prostate cancer is the second most common cancer in men and the fifth leading cause of cancer death worldwide [1]. The accurate and timely detection of prostate cancer can greatly reduce the mortality rate of prostate cancer and alleviating the pain of the patients. Clinically, carrying out a needle biopsy is the standard approach to diagnose prostate cancer [2]. Prostate biopsies need to be performed by a radiologist specializing in the genitourinary tract, which is invasive and associated with some risks like bleeding and infection. As a result, it is not a preferable alternative for patients' diagnosis. To assist early detection of prostate cancer, elevated serum level of prostate-specific antigen (PSA) has become a widely used biomarker for needle biopsy. However, a substantial number of men are put at risk of being overdiagnosed and given additional treatment by the PSA screening [3], causing many patients who do not have prostate cancer to undergo repeated biopsy.

In recent years, more and more data-driven methods have been applied to the early detection of prostate cancer to help improve the accuracy. Previous studies have shown that data mining methods, such as logistic regression, decision tree, artificial neural network (ANN), and support vector machine (SVM), which employ a variety of indexes associated with diagnostic results in model training, can improve the accuracy of prostate cancer diagnosis [4-5]. With the degree of medical informationalization continuously rising, an increasing amount of clinical data are collected and stored in databases, demanding on new machine learning algorithms to improve the prediction performance. The idea of ensemble was proposed based on the fact that different base learners have different prediction errors and the prediction accuracy can be improved by combining different learners. It is well recognized that an ensemble classifier is able to outperform a single classifier with respect of accuracy and robustness in various research fields [6]. However, most of the existing ensemble methods concentrate on the improvement of accuracy but overlook the interpretability of prediction models.

In order to construct a diagnostic model for prostate cancer with both high accuracy and good interpretability, this paper proposes a stacking-based decision tree ensemble method in a multi-objective optimization framework that can produce diagnostic rules. This method combines two level models for an accurate prediction: one is the base learners for preliminarily predicting the posteriori class probabilities of samples and the other is a meta-learner for predicting the final class label by combining the base learners. During the process of model construction, model selection and model combination are explored from a new perspective. In model selection, we take advantage of multi-objective evolutionary algorithm to optimize the accuracy and ensemble complexity simultaneously for an accurate and understandable ensemble. In model combination, we employ random forest as stacking method to find out the best way of combining the results of base learners, which can make better use of label information from base learners. To the best of our knowledge, random forest has not been utilized as stacking classifier in ensemble methods in existing studies. The performance of our method is evaluated by the prostate cancer detection experiments using various quality measures including accuracy, specificity and sensitivity. Compared with three ensemble methods (AdaBoost, random forest and gradient boosting) and four single learning algorithms (decision tree, linear-SVM which uses a linear function as kernel function, RBF-SVM which uses radial basis function (RBF) as kernel function and RBF neural network), the proposed method shows better performance. Furthermore, the diagnostic reference process is uncovered by extracting meaningful interpretable rules, which are proven to be accurate in prostate cancer detection, from the constructed prediction model through a rule extraction procedure.

The remainder of this paper is organized as follows. Section II reviews related work on early detection methods for prostate cancer and decision tree-based ensemble methods. Section III describes the problem and proposes the solution method. Section IV presents experimental results and discussions. The last section concludes the paper and suggests some topics for future research.

2 Literature review

This paper introduces a new decision tree-based ensemble learning method for prostate cancer detection. In this section, we review the literature on the following two aspects: detection methods for prostate cancer and decision tree-based ensemble learning methods.

2.1 Prostate cancer detection methods

A number of methods have been developed for prostate cancer screening. One of the early detection testing is PSA testing. Thompson *et al*. [7] used various cut-offs of PSA levels for detecting prostate cancer and found that PSA at 3.1 ng/ml showed a sensitivity of about 0.32 and a specificity of about 0.87. Some experts recommend a screening with digital rectal examination (DRE) in conjunction with PSA. In the study of exploring the specific role of DRE in detection of prostate cancer, it is revealed that patients with a positive DRE were more likely to have cancer among the ones with a PSA>3 ng/ml [8]. But it is believed that DRE as a stand-alone test is not appropriate for prostate cancer detection [9].

Many other factors such as age, race and family history are suggested to be considered when making the option of biopsy as well as DRE and PSA [10]. Some calculators are developed to help determine whether biopsy is indicated by combing various clinical variables, such as Prostate Cancer Prevention Trial risk calculator (PCPTRC) [11] and P (biop+) [12], in which logistic regression models were used. A variety of machine learning methods other than logistic regression have been applied in prostate cancer detection, such as decision tree, ANN, SVM, among many others. Bermejo *et al.* [5] employed decision tree and logistic regression to construct diagnosis models for making a distinction between prostate cancer and benign prostatic hyperplasia patients. Çinar *et al.* [13] used a reduced dataset, which contains four features (PSA, prostate volume, density, and smoking), to train an ANN and SVM for early prostate cancer detection respectively. Sung *et al.* [14] developed a support vector machines-based computer-aided diagnosis (CAD) system for prostate cancer detection which achieved an accuracy of 83%, a sensitivity of 77% and a specificity of 77%.

In recent years, ensemble learning techniques have gained increasing attention because of their effectiveness in achieving higher accuracy with good generalization ability in comparison with single classifier. Ensemble learning solves problems by generating multiple models and combining them to obtain the final prediction results. Albashish *et al.* [15] proposed a new multi-scoring features selection method based on support vector machine-recursive feature elimination (SVM-RFE) for prostate cancer detection. Recently, Xiao *et al.* [16] applied random forest to prostate cancer detection by generating a model considering the variables of transrectal ultrasound findings, age, and serum PSA levels. The experiments verified that the proposed model could provide a more accurate approach to decide whether invasive biopsy is necessary.

Among the popular algorithms applied in prostate cancer detection, ANN, SVM and ensemble methods based on them can achieve good classification accuracy, but they do not have interpretable ability to explain the interactions among the features which decide the final predictions. Although logistic regression and decision tree are self-explainable, the accuracies of both methods are not desirable. Decision tree uses predictive variable as tree node and creates new branch by computing the error reduction brought by the node split value, that is to say, the route from the root node to any leaf node can be extracted as an interpretable rule. Therefore, we choose decision tree as the base learner for constructing an ensemble with good interpretability.

2.2 Tree ensemble learning methods

Ensemble learning methods are a type of state-of-the-art learning methods, which construct multiple learners (which are also called base learners or individual learners) and combine them using some strategies. Here, we focus on the description of the related work on ensemble methods whose base learner is a decision tree.

Traditional ensemble methods create base learners by manipulating features or samples and aggregate all the base learners by a combination strategy like voting. Bootstrap aggregating ("bagging" for short) [17] uses bootstrapped copies of the training data to generate distinct classifiers. Random forest [18] is one of the methods derived from bagging, which trains different decision trees using both bootstrapped samples and the randomly generated feature subsets for each split node choosing. Ho [19] proposed the random subspace method to construct a decision forest where each tree is trained based on a random feature subset which is called a random subspace. Freund [20] introduced a new boosting algorithm called AdaBoost which associates the samples that were not classified correctly with an increased weight for the next learning to handle the hard-to-classify samples. In addition, there are many other modified ensemble learning methods similar to the aforementioned algorithms [21-22]. Although a plenty of ensemble learning methods have been proposed for all kinds of applications, it is still a challenging task to develop a highly performing and understandable ensemble system for prostate cancer detection. The ensemble approaches mentioned above still have limitations in classification performance due to class imbalance of cancer samples. On the other hand, these approaches only provide the classification results while the inference processes remain unknown.

In order to further improve the performance of ensemble, "selective ensemble was proposed to carried out model selection instead of combining all the classifiers generated which has been demonstrated to be effective in helping reduce model complexity and improve computation efficiency with a better classification performance [23]. One of the model pruning methods is to select a subset of base classifers from a ranked ensmble member list which contains the original classifiers after sorting according to the predefined criteria [24]. Another way is to use a clustering algorithm to divide the ensemble member into several clusters based on the similarity among them and then select the most accurate one from each cluster, which makes sure that the selected base classifiers are diverse and accurate [25]. The last approach is to consider model selection as a combinatorial optimization problem to find out a subset of members that can compose a better ensemble [23]. In this paper, we perform model selection with the third approach using multi-objective evolutionary algorithms (MOEAs). Among the existing research using MOEAs, most of them aim to find best sub-ensemble by optimising accuracy and diversity which are highly related to the perforamce of ensembles [26-27]. Unlike these ideas, we here dedicate to finding out the sub-ensemble with the best accuracy and minimum ensemble complexity as we aim to construct an accurate and understandable model.

Model combination plays a critical role in ensemble model construction. Among the existing ensemble methods, most of them directly turn the original inputs into output predictions using majority voting or its variants, which is merely a one-level prediction [28]. Unlike the tree ensemble methods mentioned before, this paper combines base learners by stacking, in which a meta-learner (meta-classifier) is used to combine the predictions of base learners which are called meta-data. Stacking is originally proposed by Wolpert [29] as “stacked generalization” which can usually enhance classification performance by extracting knowledge from meta-data. However, stacked generalization is time-consuming due to the fact that one usually relies on V-fold cross-validation procedures to estimate the posteriori class probabilities for each sample which are served as input for the meta-classifier, so as many other stacking approaches [6, 30-31]. In contrast, we design a time-saving stacking approach without performance loss, in which decision tree for regression (regression tree) is chosen as the base learner for generating the posteriori class probabilities like using ANN and SVM [32]. Several training strategies have been proposed to exploit the label information in the meta-data，such as multi-response linear regression (MLR) method [33], decision template [6], and decision tree [34]. Here, we employ random forest to learn the relationship between the predictions of base learners and the true class, which have shown excellent prediction performance in all kinds of machine learning tasks.

Rule extraction needs to be performed to understand the inference process of the constructed model. Sirikulviriya *et al*. [35] proposed a rule integration method to improve the comprehensiveness of rules derived from multiple trees. Mashayekhi *et al*. [36] computed the scores of rules obtained form decision tree ensemble based on rule accuracy and coverage for rule selection. Lu *et al*. [37] carried out rule refinement and rule extraction to obtain a final rule set with high accuracy and acceptable coverage. Here, we take accuracy and the generalized interpretability which reflects rule coverage and rule complexity into account to extract interpretable rules.

To summarize, in view of the incomprehensibility of the existing black box model for prostate cancer detection, we propose to use regression tree as the base learner of the ensemble classification system and to prune the original ensemble by optimizing the classification accuracy and ensemble complexity simultaneously with a multi-objective evolutionary algorithm. At the same time, in order to obtain better integrated results in combination phase and acquire the importance of the sub-models where diagnostic rules come from, we propose to employ random forest as stacking method for sub-model combination. After the prediction model is constructed, decision rules are extracted from the model. Finally, the prediction accuracy of the constructed diagnostic model is tested on a real-world prostate cancer dataset. The performance of the extracted diagnostic rules is evaluated with the score of generalized interpretability and the accuracy.

3 Problem Description and Methodology

Prostate cancer screening refers to employing simple and effective examination methods to detect tumors before the patient has evident symptoms. The early screening of prostate cancer mainly relies on PSA examination at present. However, there are a lot of controversies about the use of PSA in detection of prostate cancer because it can easily lead to wrong diagnosis. There are some other potential risk factors that can result from the routine examinations for cancer detection, such as blood routine and urine routine. The relationship between these factors and prostate cancer needs to be explored.

Clinically, the examination results are the primary reference data for doctors to identify the illness of patients based on their knowledge and medical experiences. However, this correlation is hard to find out by doctors when the number of variables is huge. From the perspective of machine learning, prostate cancer detection can be formulated as a binary classification problem to partition patients into the type of having cancer and the type of not having cancer. Machine learning algorithms can learn from existing clinical data and make predictions on new data for patients. Thus, model construction becomes the primary task to solve the problem of cancer detection. To make the screening process understandable, rule extraction becomes the second need.

The input of the prediction model for prostate cancer detection are from the demographic information of patients and all kinds of examination results including PSA testing, ultrasound scans, blood routine examinations and urine routine examinations. The diagnostic result by pathological examination is the target variable. Given a set of inputs and their corresponding diagnostic results, a prediction model can be trained, and a number of diagnostic rules can be obtained for predicting diagnostic results of new cases. Let $\{X_1, X_2, \dots, X_\mu\}$ denote the μ predictive variables and Y denote the target variable, the relations between input and output can be formulated as:

$$Y = f(X_1, X_2, \dots, X_\mu) + \varepsilon \tag{1}$$

where f represents an unknown function which is the prostate cancer detection model we intend to train using data, ε is a random variable that stands for error or random noise in data. The prediction Y_{pre} of new case $\boldsymbol{X_{new}}$ can be obtained by: $Y_{pre} = f(\boldsymbol{X_{new}})$. The proposed method for prostate cancer detection model construction is presented in the following subsections.

3.1 The proposed ensemble learning system

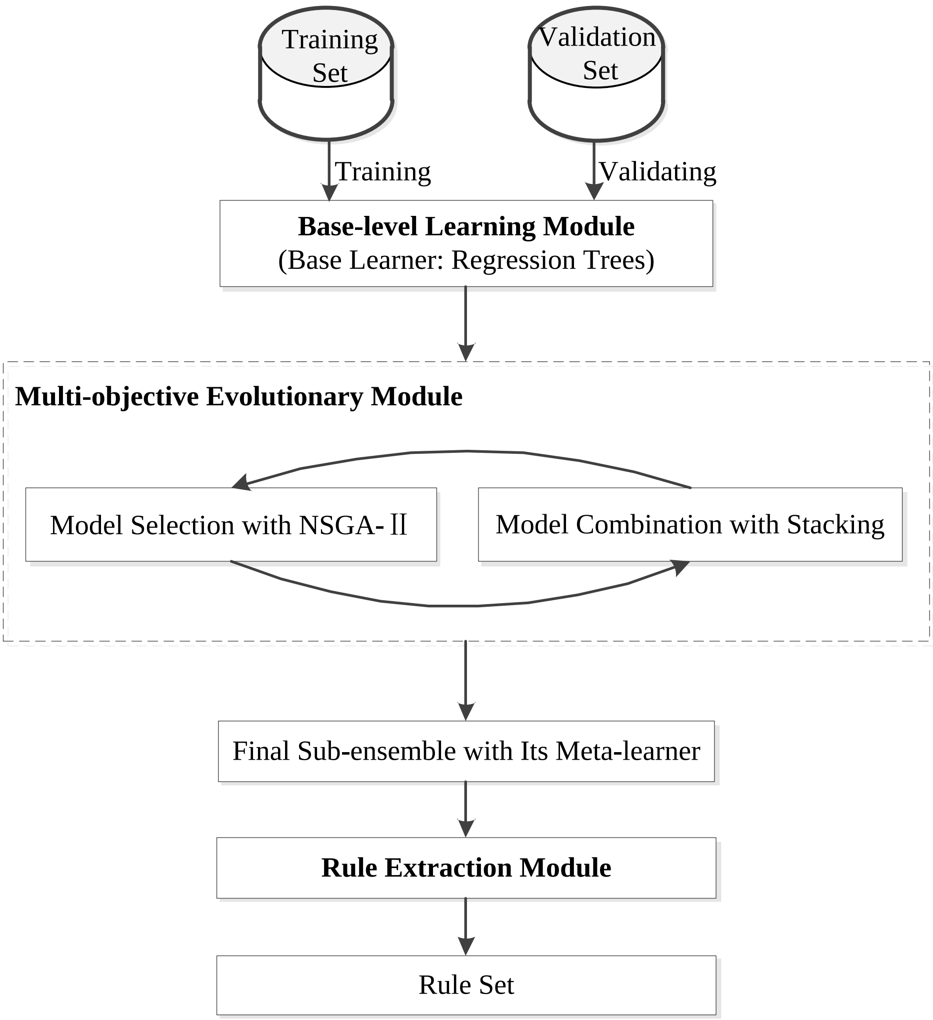

Fig. 1. Architecture of the proposed method

In this subsection, we introduce the proposed homogeneous ensemble system for prostate cancer detection. Fig. 1 illustrates the proposed ensemble system with the base learner of regression tree and the combination method of stacking, which includes three functional modules. Specially, we divide the dataset into “training set”, “validation set” and “testing set”. The first module is the base-level learning module, which consists of the base learners generated based on the training set. This module is used to generate a certain number of sufficiently diverse regression trees by manipulating the training set as the model pool for sub-ensemble selection. The second module is the multi-objective evolutionary model by which best sub-ensemble is evolved by optimizing the classification accuracy and the ensemble complexity simultaneously, where model selection is realized by NSGA-II using the validation set, and the selected sub-models are combined with a stacking method. The third module is the rule extraction module that extracts decision rules from the regression trees that make up the final sub-ensemble to support decision making by. In the following subsections, we describe each module of the ensemble system in details.

3.2 Base-level learning module

The main function of the base-level learning module is to construct base learners using the training set and to generate the meta-data for stacking model training use. In order to construct an ensemble system with both good classification performance and high interpretability, as discussed in literature review, the decision tree is our choice to construct base learners. The problem this paper deals with is prostate cancer prediction which can be treated as a binary classification task. However, if a decision tree for classification (classification tree) is utilized as the base learners, extra computation needs to be performed for obtaining the probabilities of each sample belonging to a class, resulting in a typical meta-learning paradigm. For training subset D_{n_t}, let F_{n_t} be the leaf node which samples in D_{n_t} are assigned to. In classification tree, the class label of tree node F_{n_t} is determined by the vote from samples in D_{n_t} using their real class label. The label of leaf node F_{n_t} can be computed as:

$$\arg\max_{j \in C} \sum_{i=1}^{I} c_j(x_i)\,, i = 1, \dots, I. \quad (2)$$

Where $x_i \in D_{n_t}$, C is the set of class label, I is the number of samples in D_{n_t} and $c_j(x_i)$ is the label of sample x_i. In contrast, the output of a tree node F_{n_t} in regression tree is obtained by averaging the class label of the samples in D_{n_t}, which can be calculated as:

$$\frac{1}{I} \sum_{i=1}^{I} c_j(x_i). \quad (3)$$

In the binary classification problem with class label l ($l \in \{0,1\}$), the outputs of regression tree are real numbers between 0 and 1, which stands for the posterior probabilities P_1 of samples belonging to the class with label "1". And the posterior probabilities of samples belonging to the class with label "0" is P_0 ($P_0 = 1 - P_1$). Therefore, this paper uses regression tree as the base learner of ensembles, which can produce class-conditional posterior probabilities directly. And the popular algorithm classification and regression tree [38] with the splitting criterion of LS (least squares) is used as the regression tree learning algorithm.

It is widely recognized that diversity which measures the difference of the base learners is one of the significant factors for a good ensemble. With the intention of obtaining highly discriminative meta-data for classification, the learned models should be as diverse and as complementary as possible. Consequently, it is an important task for base-level learning module to generate a large number of diverse trees. Diverse training subsets create diverse models, which can be produced by manipulating features and samples simultaneously. We use the random feature selection technique of the random subspace method proposed by Ho [19] and bootstrap sampling to generate a training subset before training a regression tree. In random subspace method, parts of features are randomly selected from the given feature space. In this paper, each regression tree is fully split using a training subset after bootstrap sampling based on the feature subspace it corresponds to. An unlabeled sample should be projected to the same subspace with the regression tree before using a regression tree.

Meta-data generation is the second task of the base-level learning module. The meta-data of traditional stacking method [6, 29, 31] is generated by a V-fold cross-validation process in which the original training set D_{tr} is partitioned into V disjoint subsets with same size. Each base learner C_k^v ($\forall\ k = 1, \dots, K$; $\forall\ v = 1, \dots, V$) is trained by a sub-training set $D_{tr} \backslash S_v$ and reserving S_v for testing, where K is the number of base learners. Thus, $\forall \boldsymbol{x_u} \in S_v$, the probability of a sample $\boldsymbol{x_u}$ belonging to a class c_m $(m = 1, \dots, M)$ is: $C_k^v(\boldsymbol{x_u}) = p_k^v(\boldsymbol{x_u}) = (p_k^v(c_1|\boldsymbol{x_u}),\ p_k^v(c_2|\boldsymbol{x_u}),\ \dots,\ p_k^v(c_M|\boldsymbol{x_u}))$. By this way, each base learner is trained by different training subset and the outputs of base learners which are used for meta-learner training are ensured to be test results for avoiding overfitting. However, this approach makes stacking become a time-consuming technique for model combination regardless of how much performance improvement it can bring. In this paper, we train diverse regression trees based on different training subset obtained by bootstrap sampling. Unlike the existing stacking methods, we employ these trees to predict the outputs of all the training samples including both those used in training and the remaining ones, which can cover more information from samples.

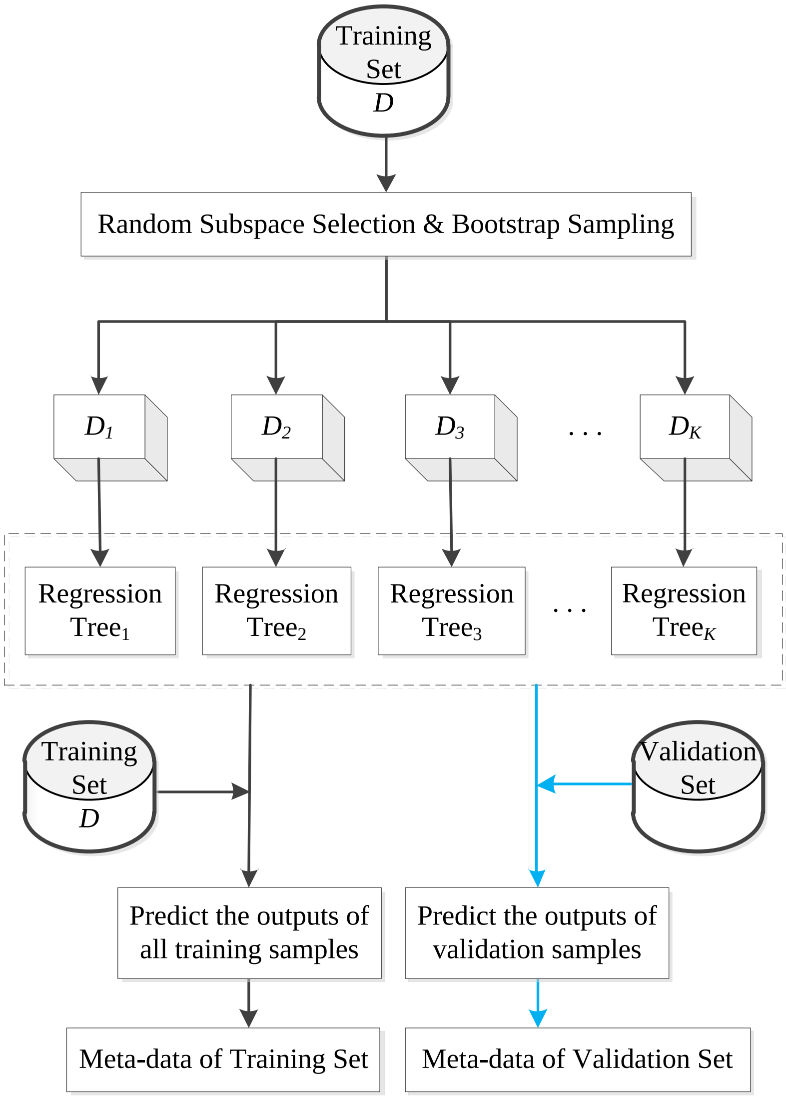

Fig. 2. Generation process of meta-data

The generation process of meta-data, including meta-data of training set (showed by black arrows) and that of validation set (showed by blue arrows), is shown in Fig. 2, where D_i denotes the *i*-th training subset after bootstrap sampling and random feature selection, *K* is the number of regression trees to generate. Once diverse base regressors are generated by training regression trees using the training subsets, meta-data of training samples can be obtained by predicting the outputs of all training samples. Meanwhile, each base regressor is validated by the validation set and the prediction results are the meta-data of the validation set.

For a binary classification problem with N_{tr} training samples, let $\boldsymbol{x_i}$ be an observation of training set D, and $p_k(y_i|\boldsymbol{x_i})$ is the probability that $\boldsymbol{x_i}$ belongs to the class with label y_m $(m = 1,2)$ given by the *k*-th base regressor (*k* = 1, ..., *K*). After getting the prediction results of K base regressors on all observations in D, the meta-data of training set, which has same shape with the meta-data of validation set, is obtained in the form of a $N_{tr} \times K$ matrix $\mathbf{L}$ as:

$$\mathbf{L} := \begin{bmatrix} p_1(y_1|\boldsymbol{x_1}) & p_1(y_2|\boldsymbol{x_1}) & \cdots & p_K(y_1|\boldsymbol{x_1}) & p_K(y_2|\boldsymbol{x_1}) \\ p_1(y_1|\boldsymbol{x_2}) & p_1(y_2|\boldsymbol{x_2}) & \cdots & p_K(y_1|\boldsymbol{x_2}) & p_K(y_2|\boldsymbol{x_2}) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ p_1(y_1|\boldsymbol{x}_{N_{tr}}) & p_1(y_2|\boldsymbol{x}_{N_{tr}}) & \cdots & p_K(y_1|\boldsymbol{x}_{N_{tr}}) & p_K(y_2|\boldsymbol{x}_{N_{tr}}) \end{bmatrix}$$

3.3 Multi-objective evolutionary module

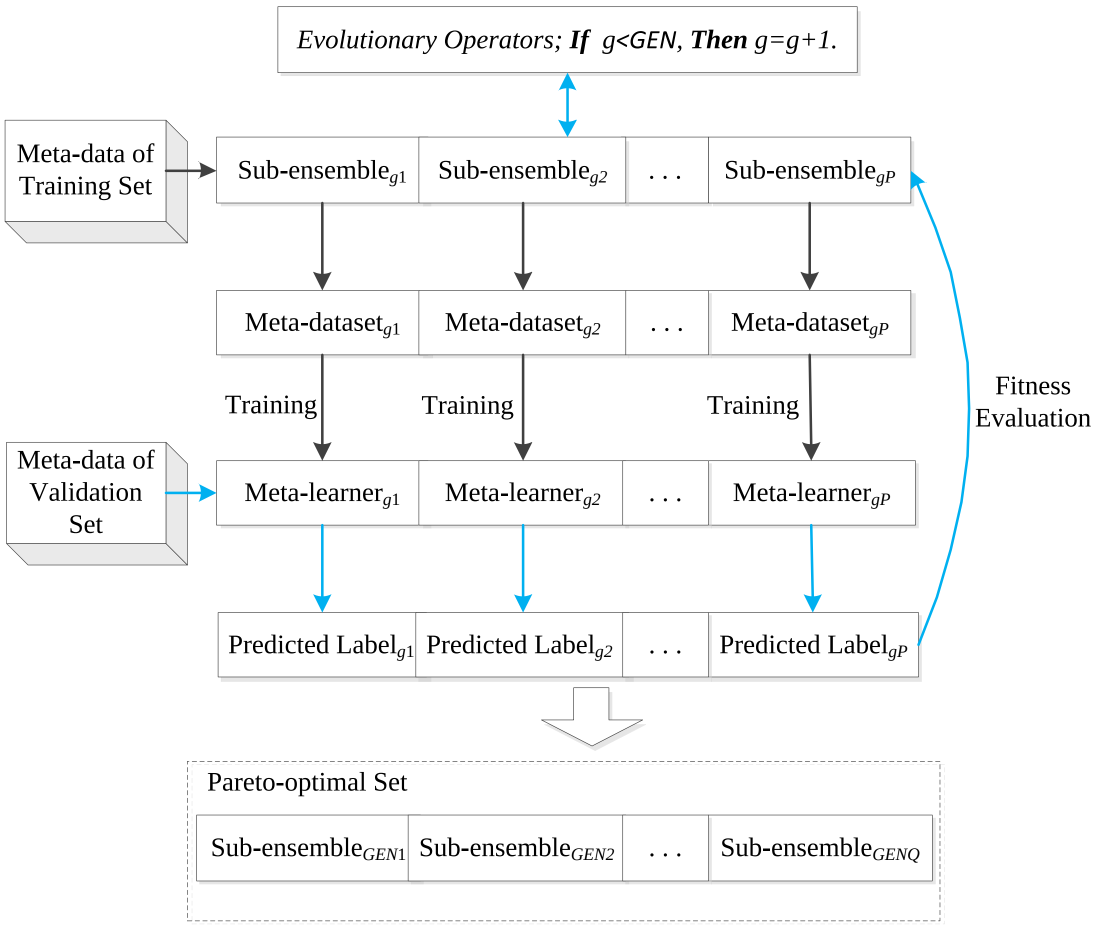

Fig. 3. Multi-objective evolutionary process

Ensemble learning methods have been demonstrated to be more effective than single learners. However, two questions always need to be concerned when constructing an ensemble: the first one is which models we should choose, and the second one is how to combine the models we have chosen. In the proposed ensemble system, the multi-objective evolutionary module depicts the techniques of model selection and model combination. Fig. 3 illustrates the evolutionary process for finding out the best sub-ensemble, where g denotes the g-th generation of evolution (g = 1, ..., GEN), GEN represents the setting number of maximum generations, P is population size of the multi-objective evolutionary algorithm, Q stands for the number of individual in the final Pareto-optimal front and the sub-ensembles correspond to the individuals of the population. In Fig. 3, the meta-data of training set is used for the construction of the candidate sub-ensembles and their meta-learners (showed by black arrows) whereas the meta-data of validation set is used for the fitness evaluation on the candidate sub-ensembles by computing the prediction labels with their trained meta-learners (showed by blue arrows). In the sequel, we focus on the description of model selection method and the model combination method in the following sections.

3.3.1 Model selection with NSGA-Ⅱ

The intention of this paper is to construct an ensemble model with good accuracy and interpretability that can generate interpretable rules. As we all know, it is not difficult to extract rules from a decision tree. However, it will be a mammoth task if we extract, hackle and integrate the rules from a complex tree ensemble consisting of a large number of decision trees. And the extracted rules may have different performance because the decision trees they belong to can achieve different accuracy. So, in order to obtain some rules with good performance, we have to reduce the ensemble complexity. The least ensemble size cannot assure good classification accuracy, nevertheless, we can choose the one with lowest complexity in the sub-ensembles having best accuracy to improve the interpretability of ensemble model.

We consider model selection as a bi-objective optimization problem to find out the sub-ensemble with the best accuracy and least base learners from the generated model pool. The optimization process dedicates to find the best combination of regression trees. Therefore, we use a binary coding scheme to encode the chromosome. Each bit of the chromosome corresponds to a trained model, and value “1” stands for a model being selected while “0” stands for a model not being selected. Thus, the length of the chromosome is equal to the number of regression trees. The two objectives of the optimization problem for model selection are formulated respectively as:

$$Maximize\ A_{gp} = 1 - \frac{1}{N_{te}} \sum_{j=1}^{N_{te}} \left[E_{gp}(\boldsymbol{x_j}) - y_j\right]^2 \quad (4)$$

$$Minimize\ C_{gp} = \sum_{k=1}^{K} Ind_{gp}(k) \quad (5)$$

where gp denotes the sub-ensemble corresponding to the p-th individual of the population in g-th generation, A_{gp} measures the accuracy of sub-ensemble gp, $E_{gp}(\boldsymbol{x_j})$ is the predicted label of the j-th testing sample by sub-ensemble gp, y_j represents the true label of the j-th testing sample, N_{te} is the number of testing samples, C_{gp} denotes the complexity of sub-ensemble gp, and $Ind_{gp}(k)$ stands for the value of bit k in the p-th individual of the population in g-th generation.

Over the past few decades, evolutionary algorithms are widely used in solving multi-objective optimization problems, which is known as multi-objective evolutionary algorithms (MOEAs). Among the great variety of MOEA methods, the elitist non-dominated sorting genetic algorithm (NSGA-II) [39] is one of the most popular and efficient one, which aims to find the non-dominated solutions under the strategy of elitism and diversity preserving. NSGA-II has been widely used in the algorithm optimization [40-41]. In this paper, we apply NSGA-II in the task of model selection which is treated as a combinational optimization problem for constructing a satisfactory ensemble. NSGA-II is a kind of multi-objective evolutionary algorithm to search for a set of Pareto optimal solutions in which none of the solutions can dominate any others, and the objective values that correspond to the Pareto-solutions form the Pareto-optimal front. Let t denote the number of generation, the procedure of model selection can be formally described as follows:

Step 1: Let t=0 and initialize the first population Pop_t;

Step 2: Decode the selected regression trees from the chromosomes in the population of the previous step to make up some sub-ensembles;

Step 3: Generate the meta-datasets that the sub-ensembles (obtained from the previous step) correspond to according to the members of the sub-ensembles and train the meta-learners using the meta-datasets;

Step 4: Predict the final predictions using the meta-data of the testing set by the trained meta-learners obtained from the previous step;

Step 5: Calculate the objective value (accuracy and ensemble complexity) and perform non-dominated sorting and crowding distance computation on Pop_t;

Step 6: Generate offspring cPop_t by implementing tournament selection, uniform crossover and 1-bit flipping mutation;

Step 7：Repeat step 2 to step 4 and calculate the objective values (accuracy and ensemble complexity) of $cPop_t$ and combine $cPop_t$ with its parent Pop_t to form $wPop_t$;

Step 8: Perform non-dominated sorting and crowding distance computation on $wPop_t$;

Step 9: Let $t = t + 1$ and create a new Pop_t;

Step 10: Repeat step 2 to step 9 until the stop criterion is met.

The main genetic operations of NSGA-II include uniform crossover, in which the attributes of two individuals are swapped with a predefined probability, bit flip mutation, in which each attribute is flipped from 0 to 1 or from 1 to 0 with a predefined probability, and selection mechanism [42] based on non-dominated sorting and crowing distance. The fast non-dominated sort is used to sort the solutions into a number of non-dominated fronts and the crowding distance [43] is then used to maintain the diversity of the individuals in the same frontier. Individuals which are located in a less crowded region (having a large crowding distance) are prioritized in selection to promote population diversity. The details of the algorithm can be found in the original paper [39]. In order to keep the individuals diverse and avoid prematurity, the duplicated individuals are removed at the end of each generation.

When the evolutionary process ends, a non-dominated solution set is obtained. Each of these solutions can be decoded to be a sub-ensemble which is a trade-off between classification accuracy and ensemble complexity. In this paper, with the purpose of constructing an accurate and understandable ensemble model, we choose the chromosome with best accuracy as the final individual, in which all the bits with value 1 corresponds to the regression trees which are selected as the members of the final sub-ensemble.

3.3.2 Model combination by stacking

Model combination is of great importance in the procedure of ensemble construction. Proper combination method can promote the prediction ability of the selected models. The most common combination method for classification is majority voting [44] and its variants, such as unanimous voting, simple majority and weighted majority voting. In this paper, we propose to utilize the stacked generalization scheme in conjunction with multi-objective optimization algorithm for constructing ensembles. According to the Wolpert's idea of stacked generalization, the outputs of an ensemble can serve as the inputs to a second-level meta-learner to learn the mapping between the outputs of the base learners and the real class label [29]. The intention of stacking is to explore a better approach to combine the trained base learners.

Numbers of stacking strategies have been proposed to predict final prediction results using meta-data. One of them assumes that each base learner makes different contributions to the final results which can be represent by a weight. In this case, the question is turned into how to find the weights. Breiman [44] employed linear stacking in regression model combination as: $y_p(\boldsymbol{x}) = \sum_{k=1}^{K} w_k h_k(\boldsymbol{x}),\ w_k \in W,\ k = 1,\ldots,K$, where $\{hkx\}$ denotes the predictions of K different models and wk is the weight of the k-th model. And the optimal weights are reckoned by carrying out a least-squares regression as: $\widehat{w} = arg\min_w \sum_{i=1}^{N}(y_i - \sum_{k=1}^{K} w_k h_k^i(\boldsymbol{x_i}))^2$, with constraint of $w_k \geq 0$, where $h_k^i(\boldsymbol{x_i})$ is the predicted value of $\boldsymbol{x_i}$ by the k-th model and y_i is the true target value of $\boldsymbol{x_i}$. Another technique is meta decision tree which is a new decision tree for model combination, in which each tree node corresponds to a classifier. What the above research tended to find is the relations between the meta-data and the final classification results. Here, the meta-data is the input of stacking model and the classification results are its output. Meta-data is produced by different base learners. Therefore, what they seek is a pattern between some features and the final label, and these features are the base learners and the meta-data is the feature values. Thus, every sample corresponds to a new observation with new features, which inspires us to employ machine learning method to learn this pattern.

In our ensemble system, random forest (RF) [18] is chosen to play the role of stacking combiner (meta-learner). RF is a popular and efficient ensemble method for both classification and regression problems. The principle of RF is to aggregate numbers of decision trees generated by randomly choosing a subset of features at each tree node using bootstrap samples. RF is proven to be computationally effective and shows good performance in amounts of diverse applications, which becomes more and more popular in solving real problems.

The predictions of the base regressors are used as a new set of features for the classification algorithm RF. These meta-data of training set are the inputs to train the meta-learner, RF classifier. For testing the current sub-ensemble or applying it to any new sample, the testing set is given to all the base regressors contained in the sub-ensemble and then their prediction vectors are given to trained RF classifier. Thus, the final predicted classes for the test data are obtained.

3.3 Rule extraction module

The rule extraction module aims to extract interpretable rules from the regression trees of the best sub-ensemble that is obtained from the multi-objective evolutionary module. One significant issue in rule extraction is the interpretability, which is also known as understandability or transparency of the generated rules [45]. Several factors are associated with the interpretability of rules, such as compactness commonly indicated by number of rules and number of premises and consistency. This paper focuses on the rules derived from decision trees which denotes as “IF $condition_1$ AND $condition_2$ AND ... AND $condition_t$, $THEN\ class$”, where t is the number of conditions. Each condition of a rule is presented in the form of “$A\ \theta\ \alpha$”, where A is a descriptive variable, α is a split point of the variable A and θ stands for “<”, “<=”, “>”,”>=” or “=”. For the rules generated from decision trees, each condition corresponds to the attribute used as a tree node. As a consequence, a rule with more conditions is more complex and less comprehensive. Therefore, we here define the complexity of a rule as the proportion of the number of the attributes the rule consists on the number of the attributes in original dataset. Besides, the coverage of a rule can reflect the interpretable ability of the rule in data, which can be defined as the percentage of the samples satisfying all the conditions of rules. In order to evaluate the rules derived from decision trees comprehensively, we take complexity and coverage as the two main aspects of generalized interpretability to estimate the rules.

At the beginning, a raw rule set can be obtained based on the selected regression trees. It is worth noticing that the input features of the trained meta-learner (RF classifier) that serves the best sub-ensemble corresponds to the outputs of the base learners of the best sub-ensemble. In RF, feature importance can be computed as the total decrement of squared error brought by that feature during the process of forest generation. Here, the feature importance values computed by RF indicate the significance of the regression trees included in the best sub-ensemble. By this way, we initialize the importance of rules from each tree.

Then it is necessary to remove the redundant conditions and rules. Redundant conditions refer to the same conditions appeared in a rule [35].

The redundant rules include covered rules and duplicate rules [37]:

(1) Covered rules: given two rules with c conditions and same attributes, in which $c-1$ conditions of the two rules are same and their prediction results are also

same, if the value range of the rest one condition of rule 2 covers that of rule 1, rule 1 is said to be covered by rule 2. For example:

Rule 1: IF $v_1 > 60\ AND\ v_2 < 1.6\ THEN\ class = yes.$

Rule 2: IF $v_1 > 70\ AND\ v_2 < 1.6\ THEN\ class = yes.$

New rule: IF $v_1 > 60\ AND\ v_2 < 1.6\ THEN\ class = yes.$

(2) Duplicate rules: if two rules are exactly the same, one of them is duplicated.

The rules being covered and the duplicate rules are removed and the remaining ones get an accumulated weight. After removing the covered and duplicate rules, the new rule set usually has fewer rules compared to the original raw rule set.

The rules are from the regression trees which are trained by a random subset obtained by bootstrap sampling and random subspace method. As a result, each tree corresponds to an out of bag (OOB) dataset with different sizes which will influence the coverage and accuracy of rules if they are computed based on the OOB data. Therefore, the coverage and the accuracy both on all samples and on the testing samples are computed. In order to rank the rules according to interpretability which is a compromise between complexity and coverage, we compute a score of all rules based on the performance about coverage on all samples, coverage on the testing samples and complexity to evaluate them. At last, we extract the best rules by computing their accuracy in diagnosis by averaging the accuracy obtained on the original dataset and on the testing set. Our paper employs a bottom-up rule extraction scheme [37] to obtain the final rule set. According to this scheme, among the sorted rules, the rules whose accuracy satisfy the user-defined threshold are retained.

Suppose there is a raw rule set R_r derived from the regression trees $\{T_{ibest} | ibest = 1, \cdots, n_t\}$, n_t is the number of regression trees for the best sub-ensemble. And the raw rule set is marked as $\left\{R_{ij} \middle| i = 1, \cdots, n_t; j = 1, \cdots m_r\right\}$, i stands for which tree the rule belongs to, j represents the ordinal number of the rule. The overall procedure of rule extraction is described in detail as follows:

Step 1: Initialize the weight of rule R_{ij} with the "feature importance" computed by the meta-learner (RF) as w_{ij};

Step 2: Rank the rules according to their weights w_{ij};

Step 3: Remove the redundant conditions of the rules;

Step 4: Remove the covered rules and update the weights of rules. For example, if R_{12} covers R_{21}, then R_{21} is deleted and the w_{12} is updated with the following equation: $w_{12} = w_{12} + w_{21}$;

Step 5: Remove the duplicate rules and update the weights of rules, the updating calculation is the same as that of the covered rules and the new rule set is $\{R_{ik}|i = 1,\cdots,n_t; k = 1,\cdots \widetilde{m_r}\}$;

Step 6: Rank the rules according to their weights w_{ik};

Step 7: Calculate the coverage coa_{ik} of rule R_{ik} on all samples according to the following formula:

$$coa_{ik} = \frac{sampleNum\_a_{ik}}{sampleNum_a} \tag{6}$$

where $sampleNum\_a_{ik}$ is the number of samples matched the conditions of rule R_{ik} in all samples and $sampleNum_a$ is the number of all samples;

Step 8: Calculate the coverage cot_{ik} of rule R_{ik} on the testing samples according to the following formula:

$$cot_{ik} = \frac{sampleNum\_t_{ik}}{sampleNum_t} \tag{7}$$

where $sampleNum\_t_{ik}$ is the number of samples matched the conditions of rule R_{ik} in testing samples and $sampleNum_t$ is the number of testing samples;

Step 9: Calculate the complexity $complexity_{ik}$ of rule R_{ik} according to the following formula:

$$complexity_{ik} = \frac{featureNum_{ik}}{featureNum_a} \tag{8}$$

where $featureNum_{ik}$ is the number of the features contained in rule R_{ik}, $featureNum_a$ is the feature number of the original dataset;

Step 10: Use min-max normalization to normalize the coa_{ik}, cot_{ik} and $complexity_{ik}$ to scale them between 0 and 1, and the normalization formula is as follows:

$$u^* = \frac{u-min}{max-min} \tag{9}$$

where max represents the maximum value and min represents the minimum value.

Step 11: Calculate the $score_{ik}$ of rule R_{ik} according to the following formula:

$$score_{ik} = w_{ik}(coa_{ik} + cot_{ik} + (1 - complexity_{ik})) \tag{10}$$

Step 12: Rank the rules according to their scores;

Step 13: Calculate the $aveAccuracy_{ik}$ of rule R_{ik} according to the following formula:

$$aveAccuracy_{ik} = \frac{1}{2}\left(\frac{correct\_a_{ik}}{sampleNum\_a_{ik}} + \frac{correct\_t_{ik}}{sampleNum\_t_{ik}}\right) \tag{11}$$

where $correct\_a_{ik}$ is the number of samples correctly classified by rule R_{ik} in all samples, $correct\_t_{ik}$ is number of samples correctly classified by rule R_{ik} in testing samples;

Step 14: Extract the rules that satisfy the pre-defined average accuracy.

In such a greedy approach, we can obtain an interpretable refined rule set from the best sub-ensemble obtained in the multi-objective evolutionary module with a high accuracy and good interpretability.

4 Experimental studies

4.1 Data preprocessing

The experiments are carried out with the prostate cancer dataset obtained from National Clinical Scientific Data Center (http://clinic.ncmi.cn) which is one of the data centers in National Scientific Data Sharing Platform for Population and Health of China. The dataset is provided by the General Hospital of PLA which contains the diagnostic records of the prostate cancer patients from the year of 2007 to 2013.

The dataset includes several excel files which contain the information of PSA testing, blood routine examinations, urine routine examinations, transrectal ultrasonography, gonadal hormone information, biochemical examinations and the diagnostic information. Here, we only use the data of PSA testing, blood routine examinations, urine routine examinations, transrectal ultrasonography which are regular examinations of prostate diseases. We remove the records that have null values and the records that have wrong values in each file. And then we recode the nominal variables whose value is "negative" with number "0". Thirdly, we join these files according to patient ID. Transrectal ultrasonography is a necessary examination to get the volume of prostate which is an important variable of patients with prostate disease. However, among the patients in the dataset, only 1657 records received transrectal ultrasonography. Finally, we get 1402 cases which have received PSA testing, blood routine examinations, urine routine examinations, transrectal ultrasonography, in which 34.69 percent patients among those who have received prostate biopsy do not have cancer in reality. We use 80 percent of all cases as training samples and the remaining are testing samples. There are 42 predictive variables totally and 1 target variable in the final data file, which are summarized in Table 1.

Table 1 List of variables used in the experiments

Variable Category	Variable ID	Variable	Variable Description
Target Variable	1	label	"0"- not prostate cancer, "1"- prostate cancer
Demographic Information	2	age	Patient age
Physical Examination	3	volume	Prostate volume
	4	pdia	The anteroposterior diameter of the prostate
Blood Routine Examination	5	B1	White blood cell count
	6	B2	Monocytes
	7	B3	Haematocrit (HCT) determination
	8	B4	Red blood cell count
	9	B5	Red blood cell volume distribution width (RDW)
	10	B6	Lymphocytes
	11	B7	Mean corpuscular volume (MCV)
	12	B8	Mean corpuscular hemoglobin (MCH)
	13	B9	Mean corpuscular hemoglobin concentration (MCHC)
	14	B10	Basophile granulocytes
	15	B11	Eosinophil count
	16	B12	Eosinophilic granulocytes
	17	B13	Hemoglobin determination
	18	B14	Platelet count
	19	B15	neutrophile granulocytes
Prostate Specific Antigen (PSA) Examination	20	tpsa	Total prostate specific antigen concentration
	21	fpsa	Free prostate specific antigen concentration
	22	fpsa/tpsa	The ratio of fpsa to tpsa
Urine Routine Examination	23	U1	Location of the urine 70% red cell's forward scattered light
	24	U2	Urine white blood cell
	25	U3	Urine white cell examination
	26	U4	Urine specific gravity determination
	27	U5	Qualitative test of bilirubin in urine
	28	U6	Qualitative test of urobilinogen
	29	U7	Qualitative test of urine protein
	30	U8	Urine red blood cell
	31	U9	Urine red cell examination
	32	U10	Urine erythrocyte forward scattered light distribution width
	33	U11	Urine yeast cells
	34	U12	Urinary epithelial cells
	35	U13	Microscopic examination of urine epithelial cells
	36	U14	Qualitative test of urine glucose
	37	U15	Urine acetone body test
	38	U16	Urine conductivity
	39	U17	Urine casts
	40	U18	Urine crystallization
	41	U19	Urine pH test
	42	U20	Urine nitrite test
	43	U21	Urine color

4.2 Classification performance indicators

In this paper, we employ three performance measures, accuracy, sensitivity and specificity, to evaluate the classification performance of the proposed method and the compared methods. Accuracy is an indicator to compute the proportion of all the correctly classified samples in all samples. Sensitivity indicates the proportion of correctly classified positive samples in all positive samples. And specificity measures the proportion of correctly classified negative samples in all negative samples.

Let TP represents the number of true positive samples, TN denotes the number of true negative samples, FP represents the number of false positive samples, and FN denotes the number of false negative samples. Then the three indicators can be calculated as follows.

$$Sensitivity = \frac{TP}{TP + FN} \tag{12}$$

$$Specificity = \frac{TN}{TN + FP} \tag{13}$$

$$Accuracy = \frac{TN + TP}{TN+TP+FN+FP} \tag{14}$$

4.3 Experimental design and parameter settings

This paper focuses on the improvement of the accuracy and interpretability of ensemble classification method. Therefore, we analyze the performance of cancer diagnostic models and then make a detail description of diagnostic rules. In order to examine the performance of the proposed method, firstly we compare it with several single classifiers that are commonly used in the existing studies for prostate cancer diagnosis, including decision tree [5], neural networks [13] and SVM [13]. Among them, RBF neural network and SVM with linear kernel function, SVM with RBF kernel function are the representatives of neural networks and SVM respectively. Secondly, we compare the proposed ensemble learning method with three other popular ensemble methods which belong to two ensemble algorithm family-bagging and boosting that perform well in prostate cancer detection [46], including random forest, AdaBoost and gradient boosting. The parameter settings are as follows.

The proposed ensemble method has two modules requiring parameter tuning, the base-level learning module and the multi-objective evolutionary module. In base-level learning module, 100 diverse regression trees are generated and the feature number of random subspace selection is chosen as 18 by extensive experiments. After adjusting the parameters through plenty of experiments, in multi-objective evolutionary module, the model selection using NSGA-Ⅱ is under the parameters that population size was 50, the maximum generation was 100, the crossover rate was 1.0 and the mutation rate was 0.05; the model combination using random forest was under the parameters that the tree number was 50 and the feature number for each node splitting choosing was 1 because the feature number for RF use corresponds to the number of base learners after decoding the chromosomes which ranges from 1 to 100.

Among the comparative methods, the decision tree was fully split based on the training set. The RBF neural network was experimented with the hidden node number of 50 after parameter tuning. In the SVM algorithm, linear function and radial basis function (RBF) are employed as kernel function respectively. The parameter setting of random forest was that the maximum number of features for split choosing was adjusted to be 30 and the maximum depth of each tree was tuned to be 5. The parameter setting of gradient boosting classification tree was that the learning rate was adjusted to be 0.5 and the maximum depth of each tree was tuned to be 4. The parameter setting of AdaBoost classification tree was that the learning rate was adjusted to be 0.3 and the maximum depth of each tree was tuned to be 6.

4.4 Results and discussions

After performing the 25 runs of the proposed method and the compared methods under the parameters set in experiment design section, two analyses are performed based on these experimental results. First, we analyze the diagnostic error of the proposed method and carry out a comparative study between the proposed method and seven competing methods. Then, the extracted rules based on a certain run are introduced and explained.

4.4.1 Diagnostic model analysis

(1) Analysis on performance of the proposed method

In order to analyze the generalization performance of the sub-ensembles on unseen data, both validation and testing results are summarized. The Pareto-optimal solutions from one run on validation set of prostate cancer data and their testing results are plotted in Fig. 4, where the ordinate stands for misclassification rate (MR) which is computed as $MR = 1 - accuracy$. In this figure, the dots tagged with the value of misclassification rate and ensemble complexity of the corresponding solutions denote the tradeoffs between accuracy and ensemble complexity on validation data and the circles denote the results on testing data. Each solution in Pareto front corresponds to a sub-ensemble obtained by trading off accuracy and ensemble complexity. From Fig. 4, we can see that solutions with a higher complexity achieve a lower misclassification rate along the Pareto front. As is well known, accuracy and complexity are two conflict objectives. However, among the Pareto-optimal sub-ensembles, the one with moderate accuracy and complexity cannot meet the demand of prediction. The proposed method intends to find an accurate and easy-to-interpret sub-ensemble for prostate cancer diagnosis by optimizing the misclassification rate and ensemble complexity. As accuracy is the first critical factor for a cancer diagnostic model, the ideal sub-ensemble is the most accurate one with a complexity as low as possible. Therefore, among the solutions of Fig. 4, solution G corresponds to the best sub-ensemble whose misclassification rate on validation set is 0.13.

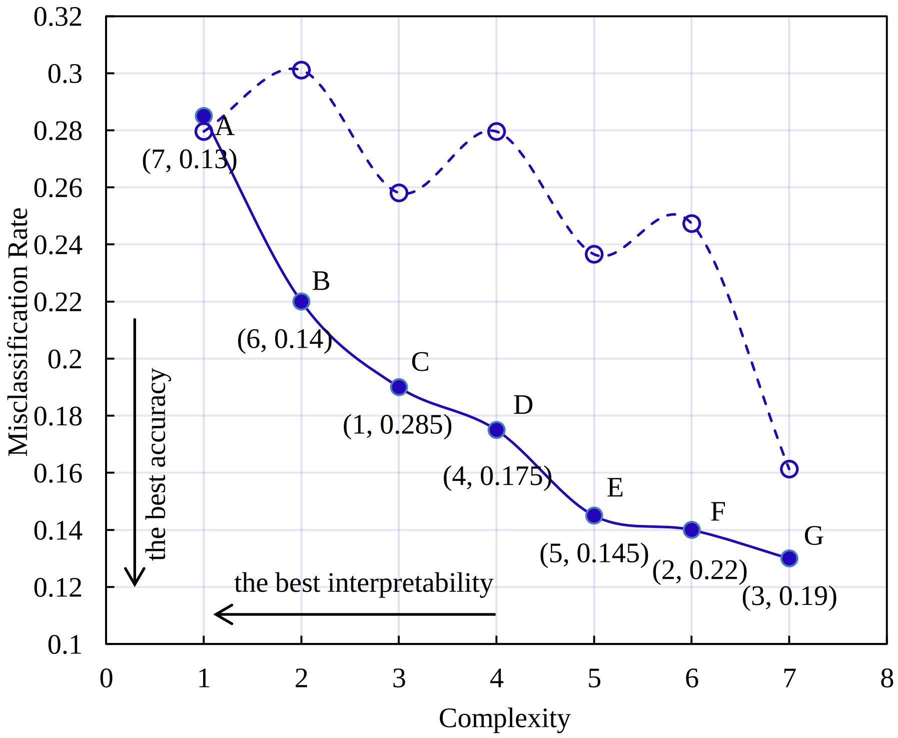

Fig. 4. Misclassification rate versus complexity of the Pareto-optimal solutions from one run. The dots denote the results on the validation data and circles test data.

The validation results and testing results of the Pareto-optimal solutions in terms of three indicators are tabulated in Table 2. This table shows that sensitivity and specificity of the solutions increases along with the increment of ensemble complexity except for that of solution B which has a lower sensitivity and higher specificity. According to the analysis on Fig. 4, solution G which has best accuracy responds to the most desirable sub-ensemble. From Table 2, it is acknowledged that the solution G also achieves the best sensitivity and specificity in both validation set and test set.

Table 2 Validation and test results of Pareto-optimal solutions from one run

Solution	Ensemble Complexity	Validation Result			Test Result		
		Accuracy	Sensitivity	Specificity	Accuracy	Sensitivity	Specificity
A	1	0.7150	0.5571	0.8000	0.7204	0.3929	0.8615
B	2	0.7800	0.5143	0.9077	0.6989	0.3214	0.8615
C	3	0.8100	0.6571	0.8615	0.7419	0.6071	0.8000
D	4	0.8250	0.6857	0.8692	0.7204	0.6071	0.7692
E	5	0.8550	0.7571	0.8846	0.7634	0.5714	0.8462
F	6	0.8600	0.7143	0.8923	0.7527	0.6429	0.8000
G	7	**0.8700**	**0.7857**	**0.9077**	**0.8387**	**0.7857**	**0.8615**

In the proposed method, an evolutionary algorithm is employed to perform model selection, in which randomness often exists in operation process. In order to test the effectiveness and stability of the proposed method, 25 prediction experiments are carried out. According to the above choosing principle of best solution, the best solution of each run is selected and the test results of these solutions are tabulated in Table 3. From this table, we can make the following observations. First, the randomness of evolutionary algorithm can lead to the random combination of individual learners and different prediction performance of ensemble models. Second, regarding the average value of 25 runs as the overall prediction performance of the proposed method, the accuracy, sensitivity and specificity are 0.8228, 0.7386 and 0.8591 respectively. Third, among the 25 runs, the lowest ensemble complexity of the sub-ensembles corresponding to the solutions is as low as 5 and the mean complexity is 9.52. Fourth, the highest accuracy the proposed method can reach is 0.8602, the sensitivity is 0.8571 and the specificity is 0.8923. In practice, a number of experiments can be carried out to look for the best ensemble model for cancer diagnosis use.

Table 3 Test results of the proposed method from 25 runs

Run Number	Ensemble Complexity	Testing Results		
		Accuracy	Sensitivity	Specificity
1	7	0.8387	0.7857	0.8615
2	11	0.8172	0.7143	0.8615
3	8	0.8602	0.8214	0.8769
4	10	0.8387	0.7143	0.8923
5	9	0.7957	0.7143	0.8308
6	7	0.8172	0.7500	0.8462
7	7	0.8065	0.7500	0.8308
8	7	0.8065	0.6429	0.8769
9	10	0.8172	0.7500	0.8462
10	7	0.8602	0.8571	0.8615
11	9	0.8495	0.7500	0.8923
12	8	0.8065	0.6786	0.8615
13	18	0.8280	0.7500	0.8615
14	11	0.8172	0.7500	0.8462
15	12	0.8387	0.7143	0.8923
16	9	0.8280	0.7500	0.8615
17	12	0.7849	0.8214	0.7692
18	8	0.8065	0.6071	0.8923
19	19	0.8172	0.6786	0.8769
20	9	0.8280	0.7857	0.8462
21	10	0.8280	0.7143	0.8769
22	6	0.8065	0.7500	0.8308
23	5	0.8065	0.6786	0.8615
24	9	0.8280	0.7500	0.8615
25	10	0.8387	0.7857	0.8615
Min	5	0.7849	0.6071	0.7692
Median	9	0.8172	0.75	0.8615
Max	19	**0.8602**	**0.8571**	**0.8923**
Ave	9.52	**0.8228**	**0.7386**	**0.8591**

(2) Comparison with other machine learning and ensemble methods

It is common knowledge that prediction accuracy is crucial to cancer diagnosis. As is analyzed in former subsections, the proposed ensemble method can achieve a low diagnostic error with a lower complexity by performing model selection using multi-objective optimization algorithm. It is well known that accuracy and interpretability are two conflict properties of prediction model. In order to get a further clarity of the prediction performance of the proposed method, a comparative study with several other machine learning methods is carried out.

First, we compare the proposed method with several commonly used single algorithms, including decision tree, SVM with linear kernel function (Linear-SVM), SVM with RBF kernel function (RBF-SVM), RBF neural network (RBFNN). The experimental results are summarized in Table 4. The results reveal that the prediction accuracy of the proposed ensemble method is better than that of the other four compared algorithms, so is sensitivity. As we all known, sensitivity indicates the classification error of the positive samples, specificity indicates the classification error of the negative samples and accuracy is a measure of classification error on all testing samples used in the experiments. Considering the imbalance of prostate cancer data, we will analyze the classification performance of these methods in term of the three indicators respectively.

Table 4 Classification results of different methods

Method	TP%	FP%	TN%	FN%	Accuracy	Sensitivity	Specificity
DT	24.07%	10.62%	54.15%	11.17%	0.7878	0.6938	0.8290
Linear-SVM	17.02%	17.66%	42.82%	22.50%	0.6044	0.4908	0.6556
RBF-SVM	0.00%	34.69%	**65.31%**	**0.00%**	0.6989	0.0000	1.0000
RBF	12.39%	22.30%	45.42%	19.90%	0.6954	0.3571	0.6954
Proposed method	**25.62%**	**9.07%**	56.11%	9.20%	**0.8228**	**0.7386**	**0.8591**

In terms of sensitivity, the proposed method obtained a highest value of 0.7386 which is followed by decision tree. Meanwhile, linear-SVM, RBF-SVM and RBFNN all show a bad sensitivity which are even less than 0.5, especially RBF-SVM with a sensitivity of 0.0, where all the positive samples are classified as negative, indicating that these patients without prostate cancer are all misdiagnosed as having prostate cancer. In this case, a lot of unnecessary medical resources and medical costs are wasted. In terms of specificity, SVM (RBF) reaches a highest value of 1.0. However, it is not a good result we need in prostate cancer detection because this "perfect" value is obtained by classifying all the samples as negative, which is at the expense of the incorrect classification of all the positive samples. Although the proposed method has a specificity of 0.8591 following RBF-SVM, here it is can be as the best classification result in terms of specificity for reason that our method can handle the influence brought by the imbalanced cancer samples. Decision tree also has a specificity of 0.8290, followed by Linear-SVM and RBFNN. The proposed method can achieve an average accuracy of 0.8228, followed by decision tree. The accuracy of Linear-SVM, RBF-SVM and RBFNN are significantly influenced by the classification error of positive samples, in which Linear-SVM has the worst accuracy. These results suggest that decision tree-based prediction models are more appropriate for prostate cancer detection.

Then, we make a comparison of the proposed tree ensemble method with a few popular tree ensemble methods, including random forest, AdaBoost and gradient boosting. According to the description of the former sub-section, the proposed ensemble method obtains an average ensemble complexity about 10. Under same complexity, we conduct a performance comparison in terms of three indicators. The parameters of the compared methods except ensemble size are optimized using validation set. The average test results of 25 runs achieved by random forest, AdaBoost and gradient boosting are summarized in Table 5. The results in Table 5 show that the proposed method performs best compared to three popular ensemble methods under same ensemble size. The model selection operation in the proposed method creates more possibility to find good ensembles with the lowest possible complexity, which does not exist in traditional ensemble methods like Random forest, gradient boosting and AdaBoost. The overall low complexity of the obtained ensemble models lays the foundations for the extraction of the interpretable diagnostic rules.

Table 5 Classification results of several ensemble methods under same complexity

Method	Accuracy	Sensitivity	Specificity
AdaBoost	0.7947	0.7208	0.8266
Gradient Boosting	0.7957	0.6786	0.8462
Random Forest	0.8065	0.7143	0.8462
Proposed Method	**0.8228**	**0.7386**	**0.8591**

4.4.2 Diagnostic rule analysis

A decision tree predicts classes of data based on a series of decision rules, which can be regarded as an integration of rules. Accordingly, an ensemble of decision trees can be considered as an ensemble of integrations of decision rules. A condensed well-performed decision tree ensemble inevitably contains the most efficient decision rules. If the rules that are critical to forecasting are visible, the decision-making mechanism of the prediction model will be clarified. This is conducive to obtaining more accurate diagnosis results combined with doctors’ medical experience.

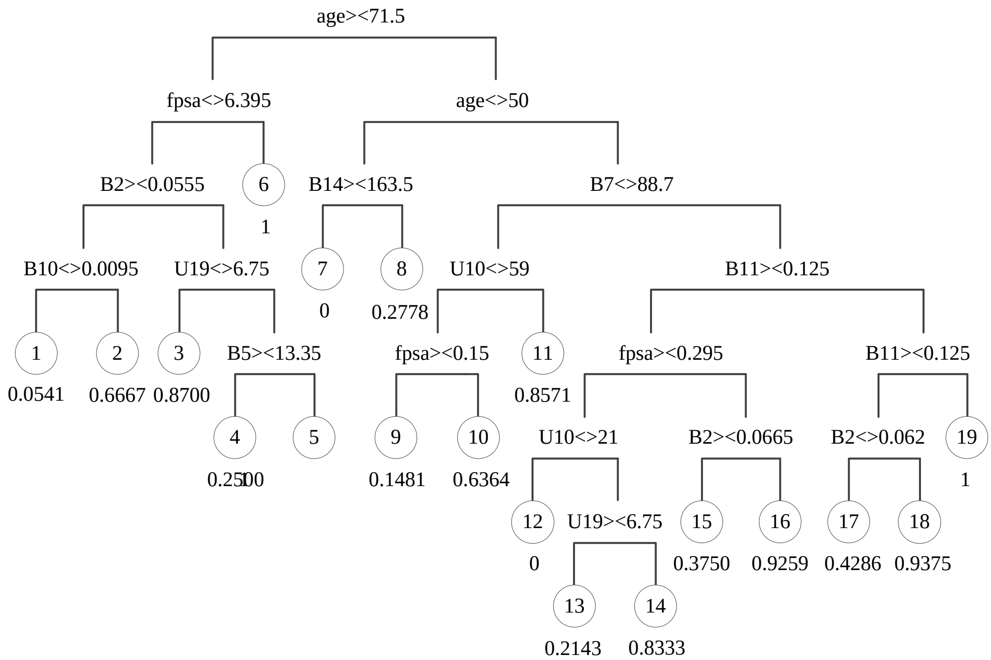

Fig. 5. Rules of one of the generated trees in the best ensemble

Taking the significance of model interpretability into account, this paper aims to construct an accurate and understandable prostate cancer detection model and extract critical decision rules. In this section, we will illustrate the diagnostic rules extracted from the best ensemble model. It is worthy of noticing that the results below are from run 2, one of the 25 runs, where we get 11 regression trees in the final sub-ensemble.

Fig. 5 shows a tree from the best sub-ensemble, from which we can see that every tree node corresponds to a variable in the prostate data. The values of leaf nodes are from 0.0 to 1.0 which indicates a posterior probability of belonging to class 1. If the value of a leaf node is 0.0 which means the posterior probability of belonging to class 1 is 0.0, the branch from the root node to the leaf node is a rule with the consequence of class 0; on the contrary, the rule has the consequence of class 1.

From the 11 regression trees, 190 diagnostic rules are obtained in total. Among these rules, a majority of them have an uncertain label which is a probability belonging to label 1. In order to ensure accuracy, we only keep the rules with a certain label in rules extraction. And there are 40 rules with a certain class label of 0 or 1. According to the rule extraction procedure described in Section 3.3, after rule ranking and rule integration, we compute the classification accuracy of these rules on the entire prostate cancer data. Finally, 13 rules are extracted whose classification accuracy are higher than 90%, which are summarized in Table 6 and are also visualized in Fig. 6. The column weight in Table 6 is equal to the feature importance computed by combiner RF which can be viewed as the contribution of the tree for final class prediction. And the score and accuracy of these rules are shown in Table 7. The rules in Table 7 are ranked based on their scores which evaluate the rules taking coverage and complexity as a whole.

Table 6 Interpretable rules after ranking with accuracy higher than 0.9000

Rule ID	Tree ID	Weight	Rule
8	9	0.1432	**IF** age< 47.5 AND pdia< 2.35 AND B6>=0.3225 **THEN** label=0
36	1	0.0918	**IF** age< 50 AND B14>=163.5 **THEN** label=0
10	9	0.1432	**IF** age< 47.5 AND pdia< 2.35 AND B3< 0.3375 AND B6 < 0.3225 **THEN** label=0
17	6	0.0787	**IF** age< 77.5 AND age>=50 AND volume>=50.62 AND B7>=77 AND B12< 0.037 **THEN** label=0
5	10	0.1147	**IF** pdia>=1.95 AND B3< 0.299 AND B15>=0.68 **THEN** label=0
11	9	0.1432	**IF** age>=47.5 AND age< 71 AND pdia< 2.35 AND fpsa/tpsa>=0.2419 AND B13>=148.5 **THEN** label=0
6	10	0.1147	**IF** pdia>=1.95 AND B3< 0.299 AND B5< 14.35 AND B15< 0.68 **THEN** label=1
33	2	0.0551	**IF** pdia< 2.15 AND fpsa>=1.215 AND B4>=3.44 **THEN** label=0
1	11	0.0768	**IF** age< 76.5 AND age>=47.5 AND tpsa>=1.78 AND U10< 27 AND B13>=138.5 **THEN** label=0
18	6	0.0787	**IF** age< 77.5 AND age>=50 AND volume>=32.28 AND B7< 77 **THEN** label=1
34	2	0.0551	**IF** pdia< 2.15 AND fpsa< 1.215 AND fpsa>=0.095 AND B4< 4.705 AND B6>=0.1395 AND B13< 103.5 **THEN** label=0
29	3	0.0512	**IF** age>=82 AND volume>=12.01 AND U10< 27 AND U14< 650 AND B7< 90.3 **THEN** label=0
35	2	0.0551	**IF** pdia< 2.15 AND fpsa< 1.215 AND fpsa>=0.085 AND B4< 4.705 AND B6< 0.413 AND B6>=0.2455 AND B7>=88.95 AND B10>=0.0065 AND B13>=103.5 **THEN** label=1

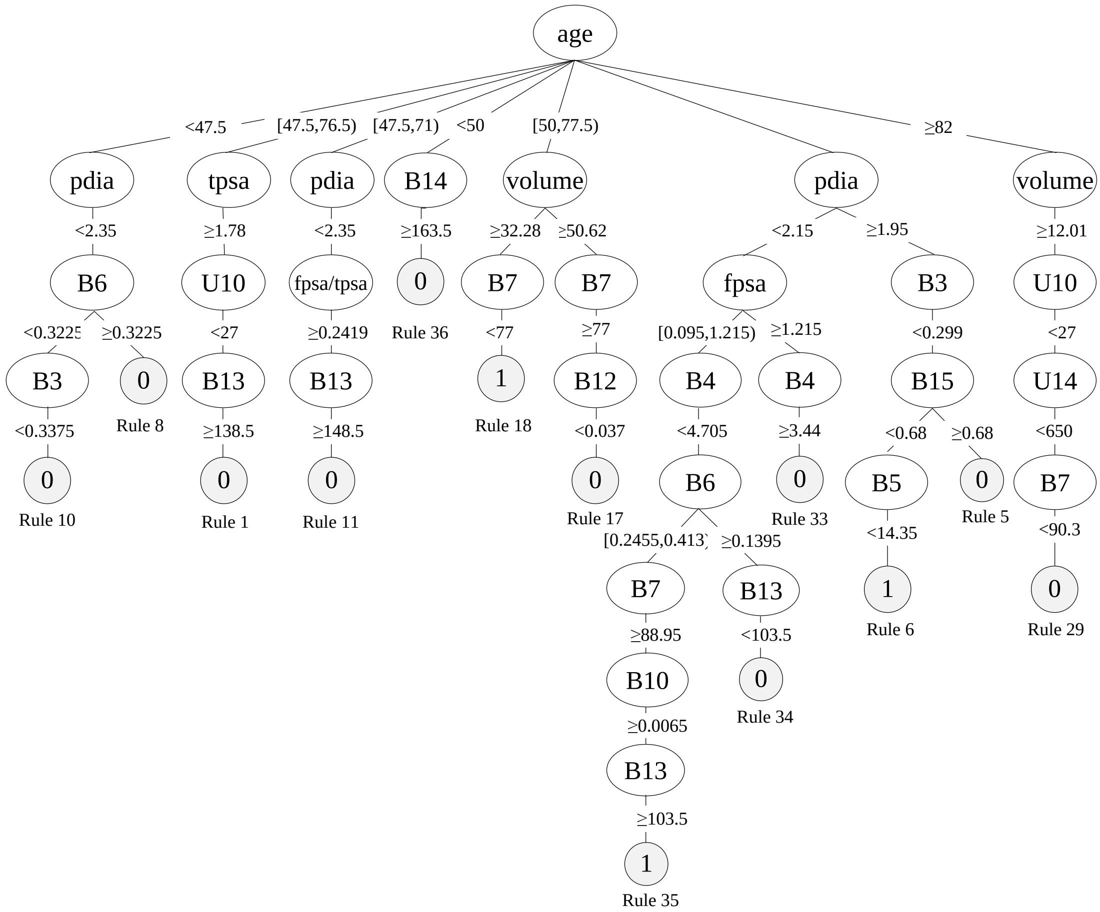

Fig. 6. Visualization of the extracted rules

Among the extracted rules, 3 rules are with class label 1, which are used to distinguish the patients with real positive results from others. And two of the rules with label 1 achieve an accuracy of 100%. Meanwhile, there are 10 rules for the diagnosis of the patients with real negative results, among which 4 rules have 100% accuracy. From Fig. 6, we can know that, “age” as one of the conditions in the 13 rules occurs 8 times and “pdia” as one of the conditions in the 13 rules also occurs 8 times, which suggest that these two variables, which represent the patient’s age at diagnosis and the anteroposterior diameter of the prostate, are two important factors in prostate cancer diagnosis. According to the NCCN (National Comprehensive Cancer Network) Clinical Practice Guidelines for prostate cancer early detection (Version 2.2017), age and PSA level are two of the most important indicators. While our experiment results show that the size of prostate and some factors in blood routine and urine routine are also closely related to prostate tumor, which needs to be validated in clinical practice. The extracted rules in Table 7 that are accurate enough can be used in practical prostate cancer detection directly. As the amount of prostate cancer data for model construction increases, more effective rules with high accuracy can be extracted by the proposed diagnostic model.

Table 7 The performace of the extracted rules

	Performance on All Data		Performance on Test Data		Comprehensive Indicators		
Rule ID	Coverage	Accuracy	Coverage	Accuracy	Complexity	Average accuracy	Score
8	0.1845	1.0000	0.1068	1.0000	0.9412	**1.0000**	0.2643
36	0.3301	0.9118	0.1748	1.0000	0.9706	0.9559	0.2490
10	0.0777	1.0000	0.0485	1.0000	0.9118	**1.0000**	0.1611
17	0.4563	0.9574	0.0680	1.0000	0.8824	0.9787	0.1543
5	0.1748	0.9444	0.0194	1.0000	0.9412	0.9722	0.1519
11	0.1553	0.8750	0.0291	1.0000	0.8824	0.9375	0.1502
6	0.0485	1.0000	0.0194	1.0000	0.9118	**1.0000**	0.1023
33	0.2233	0.8696	0.0583	1.0000	0.9412	0.9348	0.0913
1	0.1845	1.0000	0.0388	1.0000	0.8824	**1.0000**	0.0899
18	0.0194	1.0000	0.0097	1.0000	0.9118	**1.0000**	0.0606
34	0.1553	0.8125	0.0388	1.0000	0.8529	0.9063	0.0530
29	0.1068	1.0000	0.0097	1.0000	0.8824	**1.0000**	0.0423
35	0.1165	0.9167	0.0291	1.0000	0.7647	0.9583	0.0214

5 Conclusion

Over the past years, in the diagnosis of prostate cancer, PSA has been the main indicator for patients to perform a surgical biopsy, which leads to the fact that the results of a big portion of these biopsies are negative. To address this problem, we propose a stacking-based tree ensemble method for prostate cancer detection to achieve higherly accurate and well interpretable classifiers, and the information of some routine examinations of prostate cancer as well as PSA testing is used in model construction. In our method, NSGA-II is employed in model selection for finding out the sub-ensemble with best accuracy and least ensemble complexity. At the same time, random forest as meta-classifier of stacking is used in model combination. Besides, a rule extraction scheme is carried out based on the best sub-ensemble and several interpretable rules for prostate cancer detection are obtained. Extensive experiments have demonstrated the superiority of the proposed ensemble model in prostate cancer detection. Therefore, we suggest the use of understandable predictive models based on machine learning as decision support to determine the referral of patients for prostate biopsy. The shortcoming of the proposed method is that it takes more training time than single classifiers and other ensemble methods due to the model selection process, although the computational efficiency of stacking module has been enhanced. But the ensemble model trained based on our method shows a small ensemble complexity and good interpretability.

As future work, we intend to enlarge our dataset with more cases from different health centers. Thus, we can produce a more accurate self-explainable predictive model, and obtain more reliable diagnostic rules. Once models are constructed with a larger dataset, they could be extended to medical practice to help reduce unnecessary surgical procedures, thereby saving medical resources, lowering the cost of diagnosis and treatment and improving patients' satisfaction.

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